

SURVEY

Digital Twins in Sustainability Initiatives: A Review From Life Cycle Assessment Perspective

CARLOS HENRIQUE DOS SANTOS¹, ANA WOOLEY², AND ANDRÉ LUIZ ROMANO¹

¹Institute of Science and Technology, Federal University of Alfenas [Universidade Federal de Alfenas (UNIFAL)], Poços de Caldas 37715-400, Brazil

²The University of Alabama in Huntsville (UAH), Huntsville, AL 35899, USA

Corresponding author: Carlos Henrique dos Santos (carlos.henrique@unifal-mg.edu.br)

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ABSTRACT The use of Digital Twins (DTs) to aid decision-making has grown significantly in recent years, especially with the rise of Industry 4.0. Among the various goals and applications, DTs are particularly noteworthy in sustainability initiatives and Life Cycle Assessment (LCA). In LCA, DTs can replicate physical systems using real-time data, supporting various stages of the analysis. Despite its potential, there is a noted lack of theoretical foundation in this area. Therefore, this article aims to provide a systematic literature review on the use of DTs for decision-making in LCA studies. The review includes 57 studies published in scientific journals and conference proceedings available in the main scientific databases. Unlike prior reviews that have examined DTs in sustainability more broadly, this work consolidated evidence and provided a comprehensive mapping based on key research questions. We highlight that the literature on this topic is recent and growing, with applications concentrated mainly in construction and manufacturing, but also extending to supply chains, energy, mining, and consumer products. Most studies apply DTs in the Life Cycle Inventory phase to support real-time data collection via Industry 4.0 technologies, while more advanced works extend their use to impact assessment, interpretation, and scenario optimization. Overall, DTs show strong potential to enable dynamic, data-driven, and decision-oriented LCA, although challenges related to data quality, interoperability, scalability, and standardization still limit their widespread adoption.

INDEX TERMS Digital twin, industry 4.0, sustainability, life cycle assessment, systematic literature review.

I. INTRODUCTION

Digital Twin (DT) is usually described as a dynamic, virtual representation of a physical system and the processes that are related to the system, which enables the tracking and/or prediction of target variables [1]. This is accomplished with bi-directional data sent to/from the system in question. DT has been a topic of increasing interest over recent years in both academia and industry [2]. This growth is due to Industry 4.0, advances in technology, artificial intelligence (AI), wireless sensor networks, machine learning (ML), and big data [3]. These technologies provide opportunities to integrate physical processes into virtual environments.

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Sustainability is a growing topic as world powers continue to compete for resources and costs continue to rise [4]. Corporations are looking to reduce costs and reduce waste in the process and DTs have become a topic of interest in accomplishing these goals. There have been many papers and systematic literature reviews (SLR) regarding DTs and their various applications in the sustainability area. For instance, [5] focused on the use of DTs to address environmental factors of sustainability and found that DTs across the product life cycle or the DT life cycle are not sufficiently studied. Reference [6] analyzed current uses of DTs in sustainability efforts throughout the life cycle, and provided direction for future research to be done. Reference [7] reviewed several papers with an emphasis on the current use of DTs with conventional sustainability efforts and showed how the benefits

of DTs in this application are not fully utilized because there is not enough research done on this application.

Overall, existing SLRs focus on providing an overview of sustainability efforts using DTs, the limitations of existing DTs when it comes to tracking all factors associated with sustainability, and the future work to be done to expand the application of DTs in this area. While these papers and SLRs are in-depth and detailed, no SLR or other study in the literature has looked at using DT in a comprehensive Life Cycle Assessment (LCA) to support decision making and identified key features of the application. LCA is a comprehensive method to assess different impacts of a product or process over the course of its life cycle. According to [8], LCA studies have been broadly used to evaluate the environmental impacts of products, appearing as an important decision support tool within the sustainability aspects of products and processes.

Although LCA is a widely explored area of research, it is noted that the advent of Industry 4.0 provides the use of new technologies and solutions. In this case, [9] highlight the use of DTs as an important tool to support LCA studies. By mirroring the physical systems behavior, DTs can support decisions based on dynamic and continuous assessment of the current state of the modeled systems, enabling the use of real time data related to inputs and outputs of the process, making LCA more precise and automated. We note several studies that explore the use of DTs in LCA and [10] highlight that this approach not only improves the accuracy of environmental impact evaluations but also fosters a proactive, data-driven approach to sustainability. However, there is a lack of theoretical basis regarding the characteristics, practices, limitations, and future directions.

Existing reviews remain fragmented and do not systematically examine the role of DTs within LCA itself. Prior studies have explored DTs as enablers of Circular Economy strategies, as components of LCA frameworks, or as elements of broader digitalization trends in several fields. We observe that most of published reviews only address specific fields and contexts. References [11] and [12] explored the role of digitalization and DTs in LCA considering sustainable construction. [13] also carried out a review on this topic but they focused on several digital technologies, including DTs, and their role in energy management. Finally, [14] evaluated digital technologies in lifecycle management of building construction projects and, although the LCA was not the main focus, the authors mentioned this applicability.

On the other hand, in a more comprehensive way, we highlight some works that explore the role of DTs in the transition of systems from linear to the Circular Economy, such as the works proposed by [6], [15], and [16]. In this case, although the authors focused on DTs and Circular Economy as the main topics, they discussed the use of DTs in LCA-related activities. Finally, some papers addressed emerging technologies, including DTs, in LCA more directly. Reference [17] evaluate the main technologies that can be used in LCA studies, highlighting DTs as one of the main solutions adopted

by decision-makers. In the same way, [18] assessed papers that use DTs to allow dynamic LCA and [19] wrote about automation in LCA with DTs as one of the main agent in this field. Moreover, it is important to mention that, in addition to not addressing the use of DTs specifically in LCA studies, the published papers adopted different scopes regarding the period time, the selected databases, the research questions, among other characteristics. In this case, table 1 summarizes the originality of the proposed SLR and compares it to the current SRLs that address related topics.

Therefore, this work investigates the current applications and limitations of DTs in sustainability to support LCA, exploring research questions that have not been answered by other SRLs and filling an existing research gap. We reinforce that the current reviews do not identify in which processes or products DTs have been applied to support LCA, nor do they analyze at what stages of the LCA DTs are integrated. Likewise, no review provides a structured assessment of the types of DT models used in LCA or systematically discusses the advantages, limitations, technical gaps, and research challenges that arise specifically in DT and LCA integration. Addressing these omissions, the present review offers a focused and methodologically rigorous synthesis of DT applications within LCA practice.

Understanding the current applications and limitations of DTs is necessary to achieve a well-rounded and sophisticated model for sustainability to be used in LCA. According to [20], the largest hurdle to creating a DT with this application is the lack of development methods, but it does provide future work for creating a proof of concept to aid in the DT creation process. DT has the capability and potential to be used in LCA to aid in sustainability efforts. To develop a DT for this capability, it is crucial to identify and mitigate the current limitations. Thus, through an in-depth SLR, this review covers 57 studies published in scientific journals and conference proceedings indexed in leading scientific databases and, different from earlier reviews that addressed DTs in related fields, this study compiles existing evidence and presents a detailed mapping guided by research questions concerning application domains and objectives, the specific LCA stages where DTs were applied, and the key features of these technologies. Therefore, we highlight the main objectives of this paper:

- Develop an extensive systematic literature review on the use of DTs to support decisions in LCA studies.
- Identify when DT is applied to sustainability efforts over the life cycle of a product or system.
- Identify the advantages, open issues, and implementation challenges of developing DTs in LCA applications and recommend future research directions.

This work is structured as follows. Section II presents background information on concepts of the DT in sustainability and their role in supporting LCA. Section III presents the SLR methodology. Section IV presents the findings and provides a discussion. Finally, a conclusion is provided in Section V.

TABLE 1. Related reviews on the topic.

Paper	Focus	Databases	Strings	RQs
This paper	Analyze how DTs are used within LCA workflows	Scopus, ScienceDirect, Web of Science, IEEE Xplore, and SciELO (Up to June 2025)	“Digital Twin”, “Life Cycle Assessment”, “Life Cycle Analysis”, and “LCA”	General context, LCA phases, DT mechanisms, and outcomes
[15]	Role of DTs in supporting Circular Economy in construction.	Scopus, Web of Science, ScienceDirect, and Google Scholar (2017-2022)	“Circular Economy”, “digital technologies”, and related terms	Implementation, roles and barriers to DT and Circular Economy integration
[12]	Digital transformation and Circular Economy in the built environment.	Scopus, Web of Science, and Google Scholar (2010-2022)	“digital transformation”, “circular economy”, and “built environment”	Digitalization supporting Circular Economy and associated trends and barriers
[17]	Product lifecycle management applications across industrial sectors.	Scopus, Web of Science, and ScienceDirect (2008-2024)	“Product lifecycle management” and sector-specific keywords	Adoption, trends and research gaps related to Product lifecycle management
[18]	Integration of LCA, DTs and Blockchain in construction.	Scopus and Web of Science (Up to 2023)	“LCA”, “DT”, “Blockchain”, “BIM”, and Construction terms	DT and blockchain role for dynamic sustainability assessment
[11]	Digitalisation (DT, BIM) to support LCA in Asia-Pacific construction.	Scopus and Web of Science (2014-2024)	“LCA”, “LCC”, “Sustainability, Digital Twin”, and “BIM”	Trends in digitalisation for sustainability
[6]	DTs for Circular Economy in manufacturing	Scopus and Web of Science (Up to 2023)	“digital twin”, “circular” and related terms	Circular Economy challenges, DT application scenarios, and technology readiness level
[13]	Artificial Intelligence based energy systems and lifecycle perspectives	Scopus and Web of Science (2015-2025)	“Artificial Intelligence”, “Machine Learning”, and “RL” -related terms, “energy systems”, and “LCA/LCC”	Artificial Intelligence in energy management and emphasis on LCA/LCC studies
[16]	Digital technologies in LCA	Scopus and Web of Science (2013-2024)	“digital”, “LCA”, and circularity-related terms	Circular loops in LCI and trade-offs of DTs in LCA

II. DIGITAL TWINS IN SUSTAINABILITY AND THEIR ROLE IN SUPPORTING LCA

Industry 4.0, a term referring to the fourth industrial revolution, pertains to a new industrial era based on intelligent

TABLE 1. (Continued.) Related reviews on the topic.

			Terms related to	
[19]	Automation of Life Cycle Inventory generation	Web of Science (2008-2023)	LCA and automation (from a previous bibliometric analysis)	Methods for automating LCI, dominant tools, and current gaps.
[14]	Digital technologies across the construction lifecycle.	Scopus (2007-2024)	“BIM”, “GIS”, “Building”, “Construction” and related terms	BIM-GIS integration trends, applications, challenges, and future research

companies and processes. This aims to increase efficiency through the integration of real and virtual environments using emerging technologies [21]. The pillars of this new industrial era, in addition to the well-known Information Technology (IT), include solutions such as the Internet of Things (IoT), Big Data, Cloud Technology, high level of automation, AI, and Cyber-Physical System (CPS). The CPS involves the virtualization of real processes and systems through intelligent digital copies [22], [23]. In this context, when considering virtual copies of real processes, the concept of the DT emerges.

A DT can be described as a virtual replica that represents physical systems, automatically fed by real data to optimize these systems and provide responses, instructions, or commands with a certain degree of autonomy [24]. In a few words, [25] define the DT as a virtual representation that describes the behavior of a component, product, or system, including the key information needed for its evaluation throughout its entire life cycle. In a report published in 2024, the National Academies of Sciences, Engineering, and Medicine highlighted the main research gaps and future directions in DT [26]. DT is defined in the report as “a set of virtual information constructs that mimics the structure, context, and behavior of a natural, engineered, or social system (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and informs decisions that realize value. The bidirectional interaction between the virtual and the physical is central to the digital twin.”

Over the years since its inception, various definitions and key characteristics of the DT have emerged [27]. Reference [28] report that, although initially recognized as a virtual representation of a product or process, the DT is more than that. It includes algorithms capable of evaluating the real system and guiding decisions based on the system’s behavior. Reference [29] and [30] highlight the main characteristics of the DT as not only its ability to mirror the physical system in a highly synchronized and accurate manner, but also its ability to connect with various data sources and recognize the characteristics and behavior of systems over time to make increasingly efficient and precise decisions.

In contrast to decisions based on the experience and expertise of decision-makers, [24] emphasize that the era of DTs is

founded on increasingly efficient decisions that are independent of the decision-maker. Furthermore, it is important to note that, in line with the principles of Industry 4.0, manufacturing systems must be capable of monitoring and collecting information from their processes to feed the DTs, which in turn, collaborate with decision-makers [22]. Finally, according to [31], DT can support decisions through three main approaches: (i) diagnosis, to evaluate decisions based on past analyses; (ii) monitoring, to oversee and control processes; and (iii) prognosis, to anticipate and predict behaviors.

According to [32], although there are different levels of DTs, which vary based on the degree of integration between physical and virtual environments, they all rely on four main components: (i) the Physical System (PS), comprising humans, materials, and processes; (ii) the Virtual System (VS), consisting of models that represent physical behavior; (iii) the Service System (SS), which includes the infrastructure that enables bidirectional communication between the physical and virtual environments; and (iv) DT Data (DTD), a set of data and information exchanged between the systems. Figure 1 illustrates the DT structure.

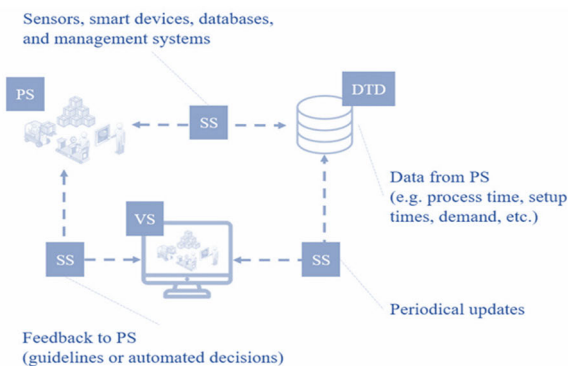


FIGURE 1. General architecture of a DT model (adapted from [32]).

Furthermore, considering the objectives of DTs, we also highlight their applicability in sustainable initiatives as a promising approach. In this way, [33] reinforced the evolution of the Industry 4.0 requires important initiatives that must consider sustainable processes and technologies such as DTs have an important role in this context. According to [34], DTs can support decisions during the entire life cycle of products and processes, involving development design, production, logistics, application, and service phases. In this case, the authors pointed out that sustainable aspects can be evaluated in each phase, supporting different decisions.

[35] highlight that sustainability in productive systems (goods and services) refers to the development of circular processes that prioritize resource reuse, recycle and remanufacturing, resulting in a comprehensive reduction of all forms of waste and positively impacting the environmental, social and economic spheres. In this way, [6] presents several examples where DTs are adopted in circular processes, where

the models support decisions related to product development, remanufacturing planning, route optimization, maintenance scheduling, among others. On the other hand, [33] highlighted that DTs also have an important role in LCA studies.

According to [36], the LCA is a useful method for evaluating the environmental impact of different products. In this case, it is possible to measure the environmental impacts across various stages of a product's life: manufacturing, usage, and end-of-life. Reference [37] reveal that LCA was proposed and improved in the 80s and 90s, representing an important methodology for evaluating the environmental and social impacts of products throughout their life cycle. Furthermore, the authors state that LCA has gradually evolved over the years from frameworks and methods that allow comprehensive and multi-objective studies. It is important to highlight that LCA is a well-structured and internationally standardized tool, as outlined in the environmental management standards "ISO 14040" and "ISO 14044".

[38] reveal that LCA has become widely accepted in recent years by both academia and industry as an important instrument for environmental decision support. In this sense, an LCA study can be divided into four phases: (i) Goal and scope definition, which involves defining the system boundary and the level of detail; (ii) Inventory Analysis, where input and output data associated with the system under study are quantified; (iii) Life Cycle Impact Assessment, where the objective is to provide additional information to help evaluate the Inventory Analysis results of a product system, aiming to better understanding of its environmental significance; and (iv) Interpretation, in which the results are summarized and discussed as a basis for conclusions, recommendations and decision-making. Figure 2 illustrates the LCA phases structure and practical applications.

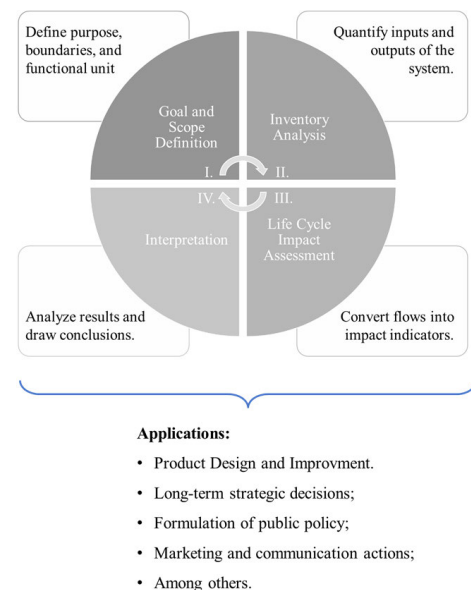


FIGURE 2. LCA phases and practical applications (adapted from [39]).

Despite the benefits of the LCA methodology, it still has significant limitations that must be addressed. Potential issues are the vast amount of data required, the accessibility of this data, and the considerable resources and time needed to conduct an LCA study [38]. On the other hand, emerging technologies brought by Industry 4.0 represent important opportunities for LCA to mitigate current issues. In this case, [10] highlight that CPS, IoT, and Simulation and Modelling techniques stand out as the most effective technologies for supporting LCA studies, allowing data collection and processing in a more efficient way compared to traditional approaches. The authors also reveal that DTs are promising solutions in LCA studies, since they transform how data is gathered and used throughout the product life cycle, significantly aiding in the creation of detailed eco-profiles.

The integration of DTs in LCA offers numerous advantages, offering a powerful means to improve sustainability practices [10]. One significant benefit is the ability to simulate various scenarios and predict the environmental impact of different design choices, materials, and processes [40]. By leveraging these simulations, companies can identify the most sustainable options before actual implementation, reducing waste, energy consumption, and overall carbon footprint. Additionally, [41] highlight that DTs enable proactive maintenance and optimization throughout the product's life cycle, ensuring that the system operates at peak efficiency, prolonging its useful life. This holistic approach not only improves environmental performance but also enhances economic and operational outcomes.

Furthermore, DTs facilitate collaboration and transparency among stakeholders involved in the life cycle of a product or system. By providing a shared virtual environment, all parties, from designers and manufacturers to operators and regulators, can access and contribute to the LCA [42]. This collaborative approach ensures that sustainability goals are aligned and that any potential issues are identified and addressed early in the process. Finally, [10] state that the main advantages regarding the adoption of DTs in LCA studies are: (i) Real-time data for precision, where DTs provide real-time data and insights throughout the entire life cycle of a product or system; (ii) Continuous monitoring and analysis, facilitating ongoing monitoring and analysis of assets or processes, aligning with the time-based perspective of life cycle thinking in LCA; (iii) Predictive modeling for informed decisions, enabling the prediction of environmental impacts resulting from design changes, material substitutions, or process optimizations in LCA; (iv) Design for Environment (DfE) Principles, where real-time environmental data guides decisions regarding materials, manufacturing processes, and product designs; (v) Holistic supply chain visibility, offering a comprehensive perspective on the entire life cycle network; and (vi) Adaptability for continuous improvement, ensuring that environmental data stays relevant and current throughout the entire life cycle.

III. RESEARCH METHOD: A SYSTEMATIC LITERATURE REVIEW

Different from a literature review, which allows exploratory research without rigorous methodological standards, a systematic literature review explores the literature in a structured and systematic way considering defined steps to answer a set of research questions [43], [44]. Reference [45] highlight that SLR is based on two fronts: (i) Thorough analysis of the primary literature within a specific field and (ii) Statistical methods designed to synthesize the findings and enhance result credibility. Hence, this paper proposes an SLR focused on investigating the use of DTs in LCA studies, following the methodology outlined by [43] and [46] based on four steps: Planning, Searching/Screening, Analysis/Synthesis, and Presentation. According to the authors, the key research objectives and questions are defined in the Planning phase. The Searching/Screening phase is focused on exploring the literature according to the defined criteria. Finally, the Analysis/Synthesis phase is dedicated to the analysis of findings and statistical procedures, while the results and main conclusions are described in the Presentation step.

A. PLANNING

To establish the objectives and Research Questions (RQs), the initial step involved conducting an exploratory search on the topic addressed in this paper. The Scopus® database, recognized as one of the largest and most comprehensive multidisciplinary databases [47], was adopted for this exploratory step. The search criteria comprised the keyword “Digital Twin” combined with the Boolean operator “AND” and the terms “Life Cycle Analysis” “OR” “Life Cycle Assessment” “OR” “LCA”. It is important to highlight that the terms “Assessment” and “Analysis” were considered since some works tend to adopt them as synonyms in this research field. Moreover, we considered peer-reviewed scientific journals and conference proceedings. This phase involved a series of meetings to explore aspects related to the use of DTs in LCA studies, drawing insights from the literature. Professors from two academic institutions, who have extensive expertise in topics related to the topic, participated in these discussions. Through this collaborative effort, researchers identified certain unresolved issues pertinent to this approach, which served as the foundation for formulating the RQs.

Therefore, the objectives of this research that will guide the literature search is to develop an extensive literature scan on the use of DTs to support decisions in LCA studies, creating a solid theoretical basis on this research field, as well as identifying future perspectives on the theme. Furthermore, once the objectives were defined, the RQs were formulated using the “CIMO” framework, which was proposed by [48]. This framework divides the RQs into four key elements: the problems and fields (Context), the application objectives (Intervention), the modeling techniques used (Mechanism), and the improvements, advantages, and main results (Out-

comes). Consequently, this paper combines a SLR with the CIMO methodology to explore the use of DTs in LCA studies. Therefore, the RQs for this study are:

RQ1: In which processes/products are DTs being used to support LCA applications (Context)?

RQ2: In which phases of LCA studies was the DT applied (Intervention)?

RQ3: What are the main characteristics of the DT models in the analyzed studies? (e.g., Simulation-based models, Machine Learning models, etc.) (Mechanism).

RQ4: What are the advantages, open issues, and challenges in future research paths for DTs in LCA applications (Outcomes)?

B. SEARCHING/SCREENING

Once the objectives and RQs of this study were defined, the searching/screening procedures were initiated. The first initial step involves selecting the databases for the literature review. Five databases were chosen to ensure broad and multidisciplinary coverage of the topic: Scopus®, Science Direct®, Web of Science®, IEEE Xplore®, and Scielo®. Next, the search strategy was developed using the keywords “Digital Twin” combined with “Life Cycle Analysis”, “Life Cycle Assessment”, and “LCA” within the title, abstract, or keywords section of the papers. In this case, Boolean logic (AND / OR) was employed to generate all possible combinations among these terms (see Figure 3). Approximately 15 searches were conducted across the selected databases considering the different combinations between the selected keywords. To ensure transparency and reproducibility, specific inclusion and exclusion criteria were applied. The cutoff date for all database queries was June 2025. The inclusion criteria required that: (i) the search terms (“Digital Twin” AND “Life Cycle Analysis” OR “Life Cycle Assessment,” OR “LCA”) appear in the article title, abstract, or keywords; (ii) the publication be a full-length article published in peer-reviewed scientific journals or peer-reviewed conference proceedings; (iii) the article be written in English; and (iv) the article present a practical application involving DT concepts in the context of LCA, rather than a review. Articles were excluded if they did not meet the SLR objectives, were duplicates, were not written in English, were review papers, conceptual discussion, editorials, book chapters, or non-peer reviewed materials, or did not report a practical implementation involving DTs and LCA.

The initial search resulted in unique 120 articles meeting the search inclusion/exclusion criteria, excluding redundancies. A screening process was then conducted on these selected papers, involving the reading of titles and abstracts to identify those aligning with the RQs and SLR scope. Articles not meeting the predefined criteria were excluded during this stage, resulting in 57 articles to be considered for a full-text review. In other words, we identified and excluded articles that, although meeting the criteria associated with the

keywords, did not present practical studies addressing the use of DTs in LCA. This more detailed analysis was only possible at this stage, that is, after reading the titles and abstracts. Figure 3 summarizes the procedures carried out during the Planning and Searching/Screening phases.

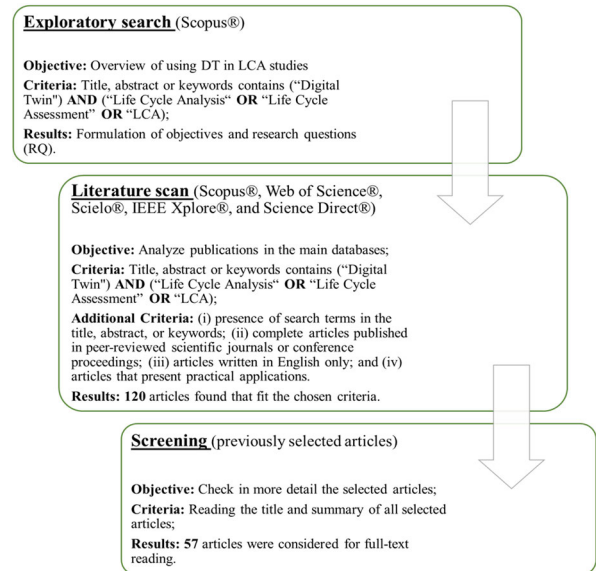


FIGURE 3. Procedures performed in the planning (exploratory search) and searching/screening stages (literature scan and screening).

C. ANALYSIS/SYNTHESIS AND PRESENTATION

The analysis and synthesis of the findings were conducted utilizing a Microsoft Excel® spreadsheet. This approach facilitated the compilation of information from all 57 reviewed articles. Each article was systematically documented in the spreadsheet according to predefined guidelines pertaining to the RQs, and the extracted results were subject to analysis using descriptive statistics. The analyses were structured around each RQ, highlighting the best practices and key perspectives concerning the utilization of DTs in LCA studies. Moreover, Section IV will present the findings, with subsections corresponding to each RQ. Tables and graphs have been employed to succinctly summarize the results and aid in their comprehension. Beyond outlining the current state of the art on the use of DTs to support decisions in LCA studies, the discussions include the advantages, limitations, and research prospects associated with the area.

IV. FINDINGS AND DISCUSSION

A. STATE OF THE ART

Among the 57 articles reviewed, it is observed that this research area is still emerging and relatively unexplored, as the earliest article examined was published in 2017. Additionally, this is a promising area of study, evidenced by the increase in annual publications. Figure 4 shows the yearly publication trends of the articles analyzed, highlight-

ing significant growth in recent years. According to the figure, we can observe that the papers were published in 24 scientific journals (65% of the papers) and 15 conference proceedings (35% of the papers). For both journal and conference proceedings, Table 2 lists the top 10 sources. The first five journals account for approximately 48.6% of the works, and the journal Sustainability stands out as the leading publication, representing about 21.6% of the works analyzed. Furthermore, most journals featured only one publication in this area, indicating that this is a multidisciplinary field compatible with various research domains and journals. In terms of conference proceedings, we identified 15 different sources, further demonstrating the field's multidisciplinary nature. In this case, we highlight the Procedia CIRP, an open-access product associated to the International Academy for Production Engineering and that publishes high quality proceedings from CIRP conferences, and the Electronics Goes Green (EGG), which focuses on sustainable solutions in the electronics and ICT sector.

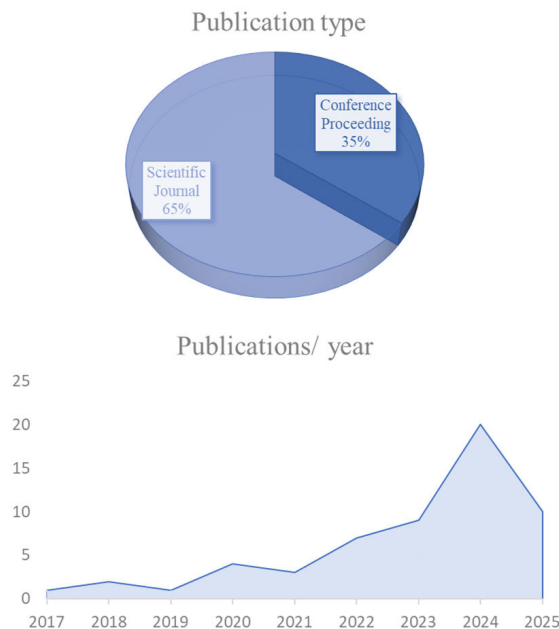


FIGURE 4. Source type and yearly publication of the analyzed papers.

B. DTS TO SUPPORT PROCESSES AND PRODUCTS' LIFE CYCLE ASSESSMENT

This section presents the findings related to RQ1 "In which processes/products are DTs being used to support LCA applications (Context)?".

There are many products and processes that DTs are used in to support LCA applications. Of the studies analyzed, the main areas of focus included the civil construction industry (24 studies, 42.1%) and the manufacturing industry (22 studies, 38.6%). Within manufacturing, the studies examined several applications such as the manufacturing

TABLE 2. Top 10 sources.

Journals	Publications	[%]
Sustainability (Switzerland)	8	21,6%
Journal of Cleaner Production	4	10,8%
Energy	2	5,4%
Journal of Building Engineering	2	5,4%
Resources, Conservation and Recycling	2	5,4%
IEEE Systems Journal	1	2,7%
Applied Sciences (Switzerland)	1	2,7%
Composites Science and Technology	1	2,7%
Computers and Industrial Engineering	1	2,7%
Digital Chemical Engineering	1	2,7%
Others	14	37,8%
Conference Proceedings	Publications	[%]
Procedia CIRP	4	20,0%
Electronics Goes Green (EGG)	3	15,0%
IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)	1	5,0%
IEEE International Conference on Environment and Electrical Engineering and IEEE Industrial and Commercial Power Systems Europe	1	5,0%
IEEE International Conference on Industrial Engineering and Engineering Management	1	5,0%
International Conference on Electrical, Communication and Computer Engineering (ICECCE)	1	5,0%
International Conference on Intelligent Systems	1	5,0%
International Conference on Engineering, Technology and Innovation	1	5,0%
IFAC-Papers Online	1	5,0%
IFIP Advances in Information and Communication Technology	1	5,0%
Others	5	25,0%

of photovoltaic cells (including their recycling) [49], [50], a woodworking section of a manufacturing process [9], home appliance manufacturing [51], extraction process design and operation [40], [52], fluid power [53], and semiconductors [54]. In the civil construction domain, the specific areas of application included the construction of general buildings, light railways [55], subway stations [56], bridges [57], [58], [59], [60], and a railway turnout [61].

A critical synthesis of these findings suggests several possible factors that help explain why construction and manufacturing dominated DT-LCA applications (80.7% combined). Both sectors have comparatively high levels of digital maturity, with widespread adoption of Building Information Modeling (BIM), Computer-Aided Design (CAD), Product Lifecycle Management (PLM), and sensor-based monitoring systems that naturally support the development of DT architectures. These established digital environments lower the technical barriers for linking operational data with LCA models. Moreover, these industries operate under increasing regulatory and market pressure to report and reduce environmental impacts, particularly related to carbon emissions, resource use, and circularity. This reinforces the need for more timely, data-driven assessment methods. Additionally, given the large environmental footprints associated with these industries, the use of DTs presents an opportunity for substantial sustainability improvements, which likely contributes to their early and concentrated adoption.

There were several categories that fell outside of these dominant areas. These categories include supply chain (4 studies, 7%), automotive (2 studies, 3.5%), geothermal energy (2 studies, 3.5%), clothing (1 study, 1.75%), mining (1 study, 1.75%), and food packaging (1 study, 1.75%). While each of these represents a smaller proportion of the sample, they demonstrate the breadth of sectors beginning to adopt DT technology for LCA-related purposes. The composition of the application of DTs in LCA applications is summarized below in Figure 5.

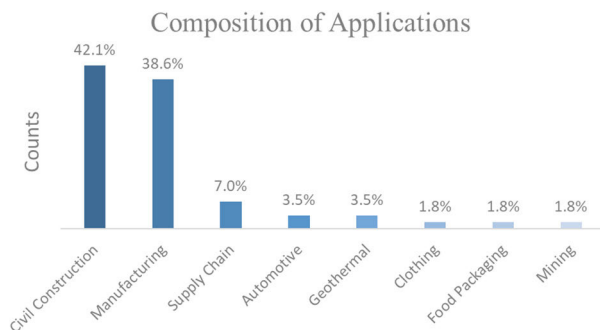


FIGURE 5. Composition of DT applications to support LCA.

The analyzed studies included specific applications within their respective processes while others provide a generalized approach to a product or process. For example, [49], [50] discussed the entire manufacturing process of mono-Silicon Photovoltaic cells including the recycling of these solar cells. At the same time, the DT in [51] was used to support the home appliance manufacturing process. Studies [9], [34], [53], [62] also discussed various manufacturing applications including, but not limited to, additive manufacturing and woodworking. Within those studies that applied a DT to the construction industry, studies [55], [57], [61] applied the DT

to railway-related construction, and studies [63], [64], [65], [66] dealt with the construction of generic buildings. Studies [56], [58], [59], [67], [68] discussed various construction applications including civil construction and subway station construction. For some of the miscellaneous products and processes that DTs are applied to, study [41] discussed the automotive industry, while studies [69], [70] were more specific by using a DT in an LCA for a geothermal power plant. Studies use a DT to support the remaining applications not discussed previously, such as mining [20], clothing [42], and food packaging [71].

C. LIFE CYCLE STAGES WHERE DTS WERE ADOPTED

This section is dedicated to answering RQ2 “In which phases of LCA studies was the DT applied (Intervention)?”. In other words, we explored the main objectives of DTs when applied in LCA studies?

First of all, it is important to note that the phases of LCA studies where DTs were adopted are dependent on the product/ process life cycle stages. In this case, [42] reveals that LCA can be developed at three moments in the life cycle of a product/process: (i) Beginning of Life (BoL); (ii) Middle of Life (MoL); and (iii) End of Life (EoL). The ISO 14040 standard also highlights the stages of a product’s life cycle that can be considered in LCA, ranging from acquisition of raw materials, manufacturing and processing, distribution and transportation, use and maintenance, disposal of waste from processes and products, recovery of used products, among other steps [39]. Furthermore, we reinforce that the stages to be considered during the life cycle depend on the area of study. For example, in the construction industry, maintenance and demolition stages can be considered [61], while clothing and automobile manufacturing industries can consider reselling stages as an important stage in their life cycle [42].

Furthermore, regardless of the product/ process life cycle stage, the papers were analyzed considering the stages of an LCA study: (I) Objectives and Scope Definition; (II) Inventory Analysis; (III) Impact Assessment; and (IV) Results Interpretation. In this case, Figure 6 summarizes the results considering the assessed papers. We note that most of the studies (35.1%) adopted DTs to support Inventory Analysis (stage II), while 31.6% of the papers went further and adopted DT in stages II, III, and IV. Finally, 7.0% of the papers demonstrated the use of DTs in all LCA stages. About 26.3% of the papers did not inform the stages or did not adopt DTs to support LCA studies.

The most frequent use of DTs in LCA studies (35.1%) is associated with the second stage of LCA (Inventory Analysis). In this case, we note that DTs are widely used to support data collection since these models are connected to physical systems and real-time/ near real-time data update them according to the physical behavior. Moreover, DTs may take advantage of Industry 4.0 solutions such as Big Data, IoT,

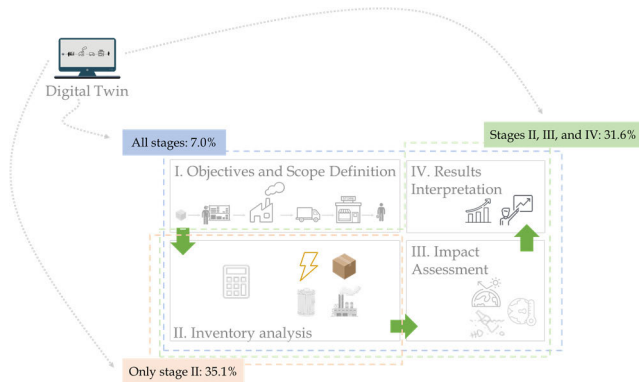


FIGURE 6. LCA stages where DTs were applied.

Cloud, and others, to improve data collection, guaranteeing quality, precision, and reliability [72]. Therefore, since Stage II of LCA studies requires data collection regarding input and output flows (i.e., resource consumption, emissions, wastes, and co-products) of each phase of the product/process life cycle, DTs might optimize this stage. These papers highlighted the role of DTs in monitor operational data, such as [66], where DT was mainly used to provide monitored energy and water use and [57] to monitor Greenhouse Gases (GHG) emissions. Reference [53] complement that several metrics can be monitored beyond to carbon footprints and related ones and, in this case, the authors call attention to the possibility to collect operational data considering the entire life cycle of products/ processes. Reference [63] carried out inventory analysis through a DT model based on the bill of materials of buildings and similar approaches can be observed in [56] and [73] where buildings/ structures were monitored through DTs and inventory analysis was supported by DT-based data. Other studies such as [65] and [74] discussed the importance of data sharing to the use of DTs in stage II, aiming robust and reliable studies. They highlighted the role of DT to allow data sharing between different LCA tools and also between different players of a supply chain. Several other examples can be observed, such as [34], [41], [75], [76], [77].

The large number of studies that adopted DTs to support Inventory analysis is not coincidental, since it is widely recognized as the most resource-intensive stage of LCA. By leveraging Industry 4.0 technologies, DTs address the central bottleneck in LCA, access to accurate, high-resolution, and continuously updated process data. This context highlights an important conceptual transition in the way LCA is being approached. In this context, while traditional LCA practices rely on static, retrospective datasets that approximate system behavior, DT-based LCAs incorporate dynamic, real/ near real-time environmental information, allowing the evolution of LCA from a backward-looking analysis tool to a forward-looking, operational decision-support system. As several of the reviewed studies demonstrate, DTs allow monitoring not only carbon emissions or energy consumption but also

material flows, equipment performance, and other operational parameters across multiple life cycle phases. This capability is essential for industries where operational variability has significant environmental implications.

On the other hand, we also observed several studies (31.6%) that went beyond regarding the use of DTs in Stage II of LCA studies. They also explored Stages III (Impact assessment) and IV (Results Interpretation). In this case, we should note that DT models were used not just for data collection, but the authors also explored the analytical ability of these models to support decisions. First, DTs might associate online/ continuous inventory results with impact categories, but exploring correlations and non-expected behaviors, an analysis that would be unfeasible in traditional LCA software. Moreover, since DTs might be based on advanced techniques and tools such as simulation, AI/ML, and optimization models, we can use DTs to support results interpretation through prediction and “what-if” analysis and scenarios optimization. Therefore, all papers that mentioned scenarios analysis and optimization were considered in this category. For example, [78] adopted DT to support inventory analysis of transportation emissions and consumptions and the authors used Geographical Information System (GIS) and a transportation model to simulate routes and material flow impact to optimize decisions. Similar approaches were proposed by [69] (for power plants), [40] (manufacturing-related process), [79] (for specific products), [80] (for buildings), among other examples. Other works can be highlighted, such as [40], [64], [81], [82], among others. These works illustrate the growing interest in utilizing DTs not solely for data acquisition but also for analytical and predictive modeling. The DTs expands the role of LCA in organizational decision-making and instead of functioning solely as an assessment mechanism, LCA studies become a platform for continuous environmental optimization, enabling stakeholders to test alternative system configurations, explore scenarios, and detect unexpected environmental behaviors and relationships. This shift signals a maturation of DT-enabled sustainability analytics, moving from monitoring toward prescriptive and anticipatory environmental management.

Finally, we call attention to a few works that fully explored the use of DTs to support LCA studies. Besides Stages II, III, and IV, these studies (7.0%) also took advantage of DTs in the Stage I (Objectives and Scope definition). This is not a trivial approach, and the authors adopted DTs set to change scopes and boundaries according to different study objectives. This is particularly interesting since it highlights the huge ability of DTs to support different kinds of activities in a flexible and user-friendly way. In this case, we highlight [9] that used DT to test different scopes and boundaries considering the entire life cycle, covering pre-manufacturing to end-of-life. Furthermore, we observed approaches mainly focused on process [70] and products [83]. We also call attention for the work proposed by [84] who tested different scenarios of

LCA considering reverse flow and recycling phases, in other words, the authors varied the scope and boundaries of LCA to evaluate best scenarios. Although still rare, the use of DTs in all LCA stages represents a major conceptual advancement. DTs allow flexible reconfiguration of boundaries, functional units, and system definitions according to different analytical objectives. This adaptability is valuable for early-stage eco-design, circularity evaluations, and strategic planning, where uncertainties about system configurations are high and multiple alternatives must be explored efficiently. These findings suggest that DTs can serve not only as computational engines but also as structuring mechanisms, guiding practitioners in designing LCAs that better reflect the complexity and variability of real-world systems.

Finally, it is important to highlight that 26.3% of the analyzed papers did not clearly mention the stage of LCA where DTs supported or did not adopt this approach as expected by the authors. First, we have works that addressed the use of DT in sustainability-related analysis during products/process life cycle, but the authors did not follow the LCA stages and, in this case, we could not associate the approach to the defined categories [20], [51], [58], [59], [85], [86], among others. Some papers addressed the topic as future perspective and did not present practical results [87] and others presented different approaches, as [42] who carried out an LCA study on DTs (although the paper fits to this review, this is an opposite approach. While DT-oriented sustainability studies are proliferating, the conceptual and methodological integration between DTs and LCA remains underdeveloped. The absence of clear reporting standards limits the reproducibility, comparability, and interpretability of DT-based LCA. Therefore, there is a pressing need for methodological guidelines, frameworks, or extensions to ISO 14040 that explicitly address digitalized and real-time LCAs.

An additional implication of the reviewed studies is the emergence of supply-chain-wide DT applications. In this context, several works highlight the potential of DTs to facilitate data sharing across different stakeholders and LCA tools. This suggests that DTs may play a pivotal role in enabling collaborative LCAs, automating emissions accounting, and supporting circularity strategies across interconnected systems. Such developments reinforce the idea that DTs are not only technological enablers but also socio-technical infrastructures capable of transforming sustainability reporting, regulatory compliance, and industrial collaboration.

The results reinforce that DT adoption in LCA aligns with a maturity gradient, from improving data quality (Stage II), to enabling deeper analytical capabilities (Stages III and IV), and ultimately to supporting strategic and exploratory design decisions (Stage I). Most industries remain at the early stages of this maturity curve, suggesting considerable potential for further integration of DT-based tools. Overall, the growing convergence between LCA and DTs represents a paradigm shift from static assessments to dynamic, data-rich, and decision-oriented environmental management, positioning

DTs as a foundational technology for next-generation sustainability analysis.

D. DTs CHARACTERISTICS

In this section, we intend to discuss RQ3 “What are the main characteristics of the DT models in the analyzed studies?”.

The analysis of the selected articles reveals that DTs applied to LCA studies rely on a wide diversity of technological bases and platforms. This diversity reflects both the variety of application contexts and different levels of conceptual and computational maturity. This finding is consistent with [88] and [89], who argue that DTs may be built upon different platforms, ranging from relatively simple visual representations to advanced analytical tools such as simulation, optimization, forecasting, and artificial intelligence (AI), among others. Based on the review of the analyzed studies, five main categories of DT technological bases were identified: (i) Simulation-based DTs; (ii) BIM/GIS-based DTs; (iii) AI/ML-based DTs; (iv) Informational DTs; and (v) Hybrid DTs, which combine more than one technological base.

Simulation-based DTs represent one of the most traditional and consolidated categories. In these studies, the core of the DT is a mathematical or computational model that represents the physical behavior of industrial processes, energy systems, structures, or transportation systems. This approach is observed, for example, in the articles [49], [50], [59], [61], [67], [69], [90], among others. In these cases, the simulation model is mainly used as inventory generators, enabling scenario analysis and environmental assessments based on the physical performance of the system. Through simulation models, decision-makers can test “what-if” scenarios and draw inferences about system behavior using deterministic and, in particular, stochastic analyses. According to [72] and [91], simulation-based DTs offer a more accessible and flexible approach, as they can be adapted to hybrid systems (automated and/or non-automated) and provide user-friendly interfaces for decision-makers. It is also noteworthy that the use of simulators as the foundation of DTs is directly linked to LCA phases in which decision-makers can move beyond static inventory analysis toward scenario evaluation and optimization of process or product parameters.

Another relevant category identified in the reviewed studies comprises BIM/GIS-based DTs, especially prevalent in construction, infrastructure, and urban system applications. Both approaches emphasize visual representations of physical systems, a characteristic that, although not mandatory for DTs, is particularly important in the context of Industry 4.0 and Cyber-Physical Systems [89]. On the one hand, GIS refers to computer-based systems designed to capture, store, analyze, and visualize georeferenced data, that is, data associated with spatial locations [1]. In this context, the use of GIS as the basis for DTs enables spatial analyses related to transportation and location, which in turn provide information

on emissions and associated environmental impacts. Consequently, LCA studies that address supply chains and broad system boundaries, such as “cradle-to-grave” and “cradle-to-gate” scopes, can significantly benefit from GIS-based DT models. Several articles adopt this approach, including [62], [68], [71], [78], among others.

On the other hand, according to [10], BIM can be defined as a method that uses a shared digital representation of a built asset to support design, construction, and operation processes. In this sense, BIM is not merely a software tool for visualizing products (e.g., buildings), but rather a process that enables more efficient collaboration among stakeholders by providing access to system components and layers. When used as the foundation of a DT, BIM-based models focus on the geometric and informational representation of buildings, assets, or territories, and are employed to extract material quantities, energy data, and spatial information that feed into LCA studies. This technological base appears predominantly in the articles [55], [57], [58], [63], [66], among others. These DTs tend to be informational or semi-dynamic in nature, placing less emphasis on detailed physical simulation, but showing strong potential for LCA applications related to embodied carbon, built stock assessment, and circular economy strategies.

AI/ML-based DTs constitute a third category, in which the DT is developed using data-driven models such as ML, Deep Learning (DL), soft sensors, or predictive models. Unlike simulation-based DTs, these approaches do not explicitly rely on physical behavior modeling, but rather on non-trivial patterns extracted from operational data. This approach is observed in the articles, such as [86] and [92] who explored DL- and Artificial Neural Networks-based models. Such DTs are commonly associated with real-time environmental monitoring and dynamic LCA applications, enabling the prediction of events as well as correlation and causal analyses across different LCA layers. These applications are mainly found in industrial and smart manufacturing contexts. AI/ML-based DTs signal a transition toward increasingly predictive and autonomous implementations. However, the reviewed studies do not explicitly address critical challenges related to explainability, algorithmic bias, and cybersecurity, which must be considered to ensure the safe and scalable adoption of AI-driven DTs, as highlighted by [72] and [88].

In addition to these approaches, a group of Informational DTs was identified, in which the DT primarily functions as a life cycle data infrastructure rather than as an advanced decision-support model. These DTs are often based on PLM platforms, Asset Administration Shells (AAS), Digital Product Passports (DPP), or structured data models [65], [74], [93]. In such cases, the DT is used to support traceability, interoperability, and circular economy initiatives by providing essential data for LCA, while the environmental assessment itself is typically conducted using external tools. Although these studies refer to their solutions as DTs, this approach closely resembles what is described as a

“Digital Shadow (DS)” by [94], since the models continuously capture information from physical systems but do not provide a complete solution capable of delivering advanced decision-support analyses without complementary tools.

Finally, the analysis becomes particularly relevant with the introduction of the Hybrid DTs category, defined as DTs explicitly built through the integration of two or more technological bases or tools. This category includes articles such as [9], who propose a framework that combines IoT devices with simulation, forecasting and optimization packages to support LCA studies. On the other hand, [77] mentioned the use of GIS models to represent physical environments combined with simulation packages to evaluate energy consumption related scenarios. Reference [78] also proposed a hybrid approach, implementing a GIS model integrated to a simulation package to evaluate supply chain networks and their impact on LCA results. Other hybrid examples can be consulted in [40], [59], [64], [69], [42], among others. Hybrid DTs may combine several techniques and tools, allowing the DT to function not merely as a representational model, but as a cyber-physical system or decision-support platform capable of evaluating different scenarios in terms of inventories and associated impacts, performing predictions, and continuously optimizing decisions. Consistently, the most advanced applications of dynamic LCA, near-real-time environmental monitoring, and continuous integration between operation and environmental assessment are concentrated in hybrid DTs. In contrast, DTs based on a single technological foundation tend to support more static or exploratory assessments. This result suggests that technological hybridization is both a trend and a functional requirement for the effective integration of DTs and LCA in complex and dynamic contexts. This finding aligns with [95], who highlights that combining multiple techniques and tools in DT models enhances their analytical capabilities and enables more effective decision-making. Figure 7 illustrates the main bases of DTs, according to the reviewed papers.

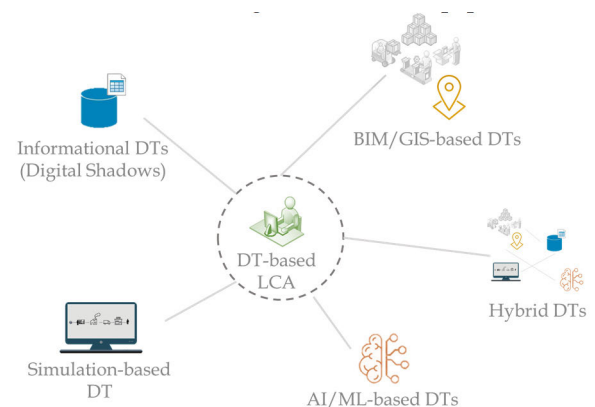


FIGURE 7. DT bases according to the reviewed papers.

Finally, we call attention to the key role of Industry 4.0 technological pillars in the DT conception. As stated

by [89] we must guarantee the continuous data collection and the full integration between physical and digital environments to take advantage of the decisions-based DTs. Thus, besides AI and Simulation, which were associated with specific categories, we highlight the importance of other technological pillars described by [96]: IoT, Cloud, Big data, blockchain, among others. Considering the reviewed papers, we observed the use of these pillars in a strategic way, as illustrated by [20], who proposed an architectural basis for DT, integrating Big Data, Cloud, IoT, and decision layers for asset lifecycle management in mining. In this way, it is possible to use IoT to collect real time data from physical systems, processing it through cloud platforms and obtain insights through Big Data. Other works also explored these pillars to compose their DTs, such as [86] who proposed a cloud-based DT as an efficient alternative to support decisions, [90] that suggested the use of IoT devices to allow real time data collection, and [18] who proposed the use of Blockchain to enhance the information flow security during the use of DTs in LCA studies. We reinforce that the use of Industry 4.0 can improve and disseminate the use of DTs in LCA studies, guarantying more efficiency and user-friendly interfaces. Therefore, further studies might explore other pillars, such as Augmented/ Virtual Reality, Robots, and Data Mining [97].

E. ADVANTAGES, OPEN ISSUES, AND CHALLENGES ASSOCIATED WITH THE USE OF DTs IN LCA STUDIES

The RQ4 “What are the advantages, open issues, and challenges in future research paths for DTs in LCA applications (Outcomes)?” will be explored in this section.

This SLR identified numerous advantages of using DT in LCA applications. One of these advantages is the ability of DTs to track all relevant factors throughout the entire life cycle of a process or product. For example, [58] proposed a comprehensive model of a bridge to monitor it throughout its life cycle. By developing a DT early in the process, all sustainability factors can be considered and a complete LCA can be addressed. Another recurring advantage lies in process monitoring and optimization. DTs can be used to track machine maintenance schedules, material degradation, and equipment lifespan, providing a basis for improved efficiency and predictive maintenance. In the construction domain, [61] further introduces the idea of 6D BIM, which incorporates more aspects relevant to construction and building life cycles such as economic and management factors. It also discusses the idea of collaboration to improve the DT to track sustainability throughout the life cycle looking at as many relevant factors as possible. Similarly, studies in manufacturing contexts [9], [50], [51], [53] demonstrate how DTs can be leveraged to evaluate multiple process iterations, optimize quality outcomes, and reduce environmental impacts. Another significant advantage of using DTs for LCA applications is their role in decision support. Researchers [34], [42], [92] emphasize how DTs provide real-time analysis of

process effects, thereby informing stakeholders and enabling more data-driven decision-making throughout the life cycle. Beyond decision-making, DTs also serve as a platform for stakeholder collaboration and communication, offering visualizations and integrated models that allow engineers, planners, and policymakers to jointly assess life cycle impacts [42], [53], [56], [61], [64]. Finally, DTs enable integration with complementary digital technologies such as BIM, GIS, and advanced simulation tools. This integration supports the development of adaptive and context-aware models that can be tailored to specific regional or industrial needs [55], [56], [57], [61], [64], [66], [73], [75].

While there are many advantages of DTs in LCA applications, there are still many open issues and challenges that limit the use of DTs in this way. One such limitation is that DTs in this application are complex and future research is suggested to develop generic methods for creating DT and maintaining them to help counter this limitation [82], [84], [90]. References [58], [62], [63], [64], and [65] look at the limitations of DTs in civil construction applications and they focus on the complexity of a DT in this application and the need for more robust data sources to create a comprehensive DT. Reference [63] specifically suggests future research in creating generic DT development methods to counter the increasing complexity of the models and supporting the creation of a comprehensive DT to track a building throughout its life cycle to see the sustainability impacts. Reference [53] analyzes the importance of future DTs managing Big Data effectively. For needs across the life cycle to be addressed effectively, clear communication networks need to be established across the process. Another promising area of future research for DTs in LCA applications is the use of DTs to predict and inform decisions. Reference [50] discuss future applications of DTs to be used in an LCA to predict sustainability factors over time. Reference [51] explores future applications within the manufacturing environment, utilizing DTs to guide business endeavors and improve the manufacturing process. By employing automatic data collection, as suggested by [9], this application of DTs could become feasible. However, further research is needed to advance this application.

Moreover, as discussed previously, managing the large amounts of data needed for DTs is challenging since we must guarantee its quality, security, timing, availability, and accuracy [18], [54], [74], [79], [80], [81], [83], [86], [98], among others. Future research in this area should include better methods for collecting and managing this data. [92] addresses the difficulty of maintaining up-to-date data throughout the life cycle and the need for more sophisticated DTs to address this issue. Reference [42] highlights the need to develop DTs for sustainability applications, emphasizing their potential to enhance sustainability across the life cycle of a product or process. Reference [34] discuss life cycle costs and the necessity for more advanced and automated methods for these calculations to ensure DTs are both worthwhile and informative.

Several research opportunities [20], [40], [56] were highlighted including development methods, restricted information exchange, and identifying target variables. Reference [20] emphasize the lack of standardized DT development methods and the need for more sophisticated approaches, including structured processes to validate and benchmark DT models before large-scale adoption. This opens opportunities for future research on developing proof-of-concept methodologies and testing their performance over the product life cycle incorporating circular-economy and user-centered principles into DT platforms [20]. Reference [56] discusses the limitations regarding information exchange and model updating, emphasizing interoperability across stakeholders and platforms (e.g., BIM, GIS, IoT) and the creation of secure data-sharing mechanisms and standard ontologies. This could cause problems when doing LCA, so the insights provided in the paper could help to mitigate these limitations and create a better DT, allowing real-time environmental and social data integration and dynamic LCA updates [56]. Finally, [40] point out the impossibility of modeling every variable in a system and suggests identifying critical variables to avoid overloading the system while still capturing the overall picture.

V. CONCLUSION

LCA studies have been evolving in the past years, mainly considering the Industry 4.0 solutions, which brought significant improvements and opportunities regarding analysis and decision-making. In this case, we highlight the use of DTs to support LCA studies as an important approach that allows more efficient and precise analysis. Therefore, this paper examined the use of DTs in LCA studies through a SLR, aiming to explore the main characteristics associated with publications in this field. We analyzed 57 articles published in scientific journals and conference proceedings, available in major literature databases. RQs were structured and addressed in the analysis sections.

Among the reviewed articles, we note that the earliest was published in 2017, indicating that this research area is still emerging and showing an upward trend in annual publications. Moreover, the publication sources indicate a multidisciplinary field. Regarding the application areas (RQ1), DTs were applied in various products and processes to support LCA applications. Among the studies analyzed, the main focus areas were the construction industry and the manufacturing industry. Other categories included supply chain, automotive, mining, food packaging, clothing, and geothermal power plants.

Moreover, we investigated the stages of LCA study where DTs were adopted considering the LCA (RQ2). In this case, among the analyzed studies, we noted that most studies focused on Stage II (Life Cycle Inventory), where DTs are mainly used to support data collection by connecting to physical systems and leveraging Industry 4.0 tools such as IoT, Big Data, and Cloud to ensure reliable and precise monitoring of physical products and processes. Beyond inventory analysis,

a significant number of studies also explored DTs in Stages III and IV, using their analytical capabilities to link continuous inventory data with impact categories, identify correlations, and perform advanced tasks such as prediction, scenario analysis, and optimization to support decision-making. A smaller group of works extended DT use to Stage I (Objectives and Scope Definition), showing how DTs can dynamically adjust boundaries and scopes to evaluate different scenarios, including reverse flows and recycling.

We also evaluated the studies with respect to Digital Twin (DT) characteristics (RQ3). The reviewed studies show that DTs applied to LCA studies rely on a wide range of technological bases, reflecting diverse application contexts and maturity levels, which can be grouped into five main categories: simulation-based, BIM/GIS-based, AI/ML-based, Informational, and Hybrid DTs. Simulation-based DTs are the most established and are widely used to generate inventories, test scenarios, and support optimization through deterministic and stochastic analyses. BIM/GIS-based DTs are especially relevant in construction, infrastructure, and urban systems, emphasizing spatial and informational representations that support material quantification, energy analysis, and supply chain assessments within LCA. AI/ML-based DTs adopt data-driven models to enable predictive and real-time environmental monitoring and optimization, signaling a shift toward more autonomous and dynamic LCA applications. Informational DTs primarily function as structured data infrastructures to enhance traceability and interoperability, often resembling digital shadows rather than full decision-support systems. The most advanced applications are found in Hybrid DTs, which integrate multiple technological bases, enabling dynamic, near/real-time environmental assessment, and continuous decision optimization. Overall, the analysis highlights the integration of Industry 4.0 pillars as key enablers for more effective, scalable, and decision-oriented DT-LCA applications in complex systems.

Finally, we explored the advantages, limitations, and future perspectives of the theme (RQ4). Among the key benefits, DTs demonstrate strong potential to provide comprehensive life cycle tracking, enable process improvements, and support more informed decision-making. By integrating data across stages ranging from material procurement and manufacturing to end-of-life, DTs can optimize production environments and contribute to more holistic sustainability assessments. Despite these benefits, significant challenges remain. The creation and maintenance of DTs demand high computational power, scalability, and considerable investment, while the scarcity and quality of data, as well as the integration of multiple databases, present further obstacles. Barriers also emerge in relation to interoperability, legal and security concerns, and restrictions on data sharing. Moreover, difficulties in modeling variables and uncertainties, combined with the limited number of studies addressing comprehensive and multi-objective assessments, including social, economic, and circularity factors, restrict the full realization of DT potential. In addition, few authors explore the integration of DTs

with complementary Industry 4.0 technologies such as AI, AR/VR, Blockchain, IoT, and simulation.

It is important to reinforce theoretical and practical contributions of this paper. This study advances the scientific understanding of DTs applied to LCA in several ways. First, it consolidates and maps the adoption of DTs across all LCA phases, providing a structured overview of how DTs support Objectives and Scope Definition, Inventory Analysis, Impact Assessment, and Results Interpretation. Second, the review contributes to the development of technical taxonomies by classifying DT architectures into simulation-based, BIM-based, GIS-based, AI-driven, customized software, and hybrid approaches, offering conceptual clarity and a reference structure for future studies. Third, it identifies methodological gaps that limit progress toward dynamic and continuous LCA, including the lack of standardized DT development frameworks, insufficient interoperability mechanisms, limited reporting of DT characteristics, and the predominance of static or manually updated DT models. Together, these insights create a structured foundation for future theoretical work in DT-enabled LCA, particularly in the design of dynamic, automated, and interoperable life-cycle assessment approaches.

The findings also offer actionable implications for practitioners, including engineers, product designers, manufacturers, sustainability professionals, and policymakers. For industry, the study clarifies which LCA phases benefit most from DT adoption, with evidence showing the strongest support occurring during Inventory Analysis, followed by scenario-based Impact Assessment and Interpretation. It also provides a clear mapping between DT tools and industrial sectors, helping practitioners select appropriate technologies (e.g., BIM for construction, process simulation in manufacturing, GIS for logistics and supply chains, AI for predictive maintenance and optimization). For decision-makers and policymakers, the study offers insights into digital environmental monitoring, highlighting how DT-enabled data streams can enhance transparency, traceability, carbon-footprint reporting, and circular-economy strategies. Collectively, these contributions support both the design and implementation of DT solutions in real operational environments, while also informing public and institutional agendas for sustainability and digital transformation.

Finally, based on the review, we emphasize that the use of DTs to support LCA studies is consistent with the ISO 14040 framework, as this approach enhances transparency, data quality, and feedback mechanisms across LCA phases, thereby aligning with ISO recommendations for robustness and reproducibility.

REFERENCES

- [1] D. Jones, C. Snider, A. Nassehi, J. Yon, and B. Hicks, "Characterising the digital twin: A systematic literature review," *CIRP J. Manuf. Sci. Technol.*, vol. 29, pp. 36–52, May 2020, doi: [10.1016/j.cirpj.2020.02.002](https://doi.org/10.1016/j.cirpj.2020.02.002).
- [2] M. Liu, S. Fang, H. Dong, and C. Xu, "Review of digital twin about concepts, technologies, and industrial applications," *J. Manuf. Syst.*, vol. 58, pp. 346–361, Jan. 2021, doi: [10.1016/j.jmsy.2020.06.017](https://doi.org/10.1016/j.jmsy.2020.06.017).
- [3] M. Syafrudin, G. Alfian, N. L. Fitriyani, and J. Rhee, "Performance analysis of IoT-based sensor, big data processing, and machine learning model for real-time monitoring system in automotive manufacturing," *Sensors*, vol. 18, no. 9, p. 2946, Sep. 2018, doi: [10.3390/s18092946](https://doi.org/10.3390/s18092946).
- [4] W. B. Adams, *Green Development: Environment and Sustainability in a Developing World*, 4th ed., Evanston, IL, USA: Routledge, 2019.
- [5] R. Carvalho and A. R. D. Silva, "Sustainability requirements of digital twin-based systems: A meta systematic literature review," *Appl. Sci.*, vol. 11, no. 12, p. 5519, Jun. 2021, doi: [10.3390/app11125519](https://doi.org/10.3390/app11125519).
- [6] J. Mügge, A. Seegrün, T.-K. Hoyer, T. Riedelsheimer, and K. Lindow, "Digital twins within the circular economy: Literature review and concept presentation," *Sustainability*, vol. 16, no. 7, p. 2748, Mar. 2024, doi: [10.3390/su16072748](https://doi.org/10.3390/su16072748).
- [7] F. Edrisi and M. S. Azari, "Digital twin for sustainability assessment and policy evaluation: A systematic literature review," in *Proc. IEEE/ACM 7th Int. Workshop Green Sustain. Softw. (GREENS)*, May 2023, pp. 1–8, doi: [10.1109/GREENS59328.2023.00007](https://doi.org/10.1109/GREENS59328.2023.00007).
- [8] O. Cavalett, M. F. Chagas, J. E. A. Seabra, and A. Bonomi, "Comparative LCA of ethanol versus gasoline in Brazil using different LCIA methods," *Int. J. Life Cycle Assessment*, vol. 18, no. 3, pp. 647–658, Mar. 2013, doi: [10.1007/s11367-012-0465-0](https://doi.org/10.1007/s11367-012-0465-0).
- [9] A. Barni, A. Fontana, S. Menato, M. Sorlini, and L. Canetta, "Exploiting the digital twin in the assessment and optimization of sustainability performances," in *Proc. Int. Conf. Intell. Syst. (IS)*, Sep. 2018, pp. 706–713, doi: [10.1109/IS.2018.8710554](https://doi.org/10.1109/IS.2018.8710554).
- [10] M. Piron, J. Wu, A. Fedele, and A. Manzardo, "Industry 4.0 and life cycle assessment: Evaluation of the technology applications as an asset for the life cycle inventory," *Sci. Total Environ.*, vol. 916, Mar. 2024, Art. no. 170263, doi: [10.1016/j.scitotenv.2024.170263](https://doi.org/10.1016/j.scitotenv.2024.170263).
- [11] E. F. Y. Liew, D. E. L. Ong, and M. I. Peerun, "Mapping the future of sustainable construction: Review on digitalisation and life cycle assessment trends in Asia-Pacific," *Energy Buildings*, vol. 350, Jan. 2026, Art. no. 116547, doi: [10.1016/j.enbuild.2025.116547](https://doi.org/10.1016/j.enbuild.2025.116547).
- [12] S. Banihashemi, S. Meskin, M. Sheikhhoshkar, S. R. Mohandes, A. Hajirasouli, and K. LeNguyen, "Circular economy in construction: The digital transformation perspective," *Cleaner Eng. Technol.*, vol. 18, Feb. 2024, Art. no. 100715, doi: [10.1016/j.clet.2023.100715](https://doi.org/10.1016/j.clet.2023.100715).
- [13] A. Ozadowicz, "Towards sustainable buildings and energy communities: AI-driven transactive energy, smart local microgrids, and life cycle integration," *Energies*, vol. 18, no. 21, p. 5668, Oct. 2025, doi: [10.3390/en18215668](https://doi.org/10.3390/en18215668).
- [14] F. H. Sarigul and H. M. Gunaydin, "Integrated BIM, GIS and interoperable digital technologies in lifecycle management of building construction projects: Systematic literature review," *Smart Sustain. Built Environ.*, pp. 1–38, Apr. 2025, doi: [10.1108/sasbe-08-2024-0312](https://doi.org/10.1108/sasbe-08-2024-0312).
- [15] X. Meng, S. Das, and J. Meng, "Integration of digital twin and circular economy in the construction industry," *Sustainability*, vol. 15, no. 17, pp. 1–14, Sep. 2023, doi: [10.3390/su151713186](https://doi.org/10.3390/su151713186).
- [16] S. Toniolo, G. Pierli, L. Bravi, L. Liberatore, and F. Murmura, "Digital technologies and circularity: Trade-offs in the development of life cycle assessment," *Int. J. Life Cycle Assessment*, vol. 2025, pp. 1–21, Feb. 2025, doi: [10.1007/s11367-025-02436-9](https://doi.org/10.1007/s11367-025-02436-9).
- [17] M. S. Aberkane and A. Otman, "Product lifecycle management: What sectors and what technologies used? A systematic literature review," *Discover Sustainability*, vol. 6, no. 1, pp. 1–17, Jul. 2025, doi: [10.1007/s43621-025-01267-w](https://doi.org/10.1007/s43621-025-01267-w).
- [18] K. Figueiredo, A. W. A. Hammam, R. Pierott, V. W. Y. Tam, and A. Haddad, "Integrating digital twin and blockchain for dynamic building life cycle sustainability assessment," *J. Building Eng.*, vol. 97, Nov. 2024, Art. no. 111018, doi: [10.1016/j.jobte.2024.111018](https://doi.org/10.1016/j.jobte.2024.111018).
- [19] B. Köck, A. Friedl, S. Serna Loaiza, W. Wukovits, and B. Mihalyi-Schneider, "Automation of life cycle assessment—A critical review of developments in the field of life cycle inventory analysis," *Sustainability*, vol. 15, no. 6, p. 5531, Mar. 2023, doi: [10.3390/su15065531](https://doi.org/10.3390/su15065531).
- [20] N. El Bazi, M. Mabrouki, O. Laayati, N. Ouhabi, H. E. Hadraoui, F.-E. Hammouch, and A. Chebak, "Generic multi-layered digital-twin-framework-enabled asset lifecycle management for the sustainable mining industry," *Sustainability*, vol. 15, no. 4, p. 3470, Feb. 2023, doi: [10.3390/su15043470](https://doi.org/10.3390/su15043470).
- [21] J. Wan, H. Cai, and K. Zhou, "Industrie 4.0: Enabling technologies," in *Proc. Int. Conf. Intell. Comput. Internet Things*, Harbin, China, Jan. 2015, pp. 135–140, doi: [10.1109/ICAOT.2015.7111555](https://doi.org/10.1109/ICAOT.2015.7111555).

- [22] R. Y. Zhong, X. Xu, E. Klotz, and S. T. Newman, "Intelligent manufacturing in the context of industry 4.0: A review," *Engineering*, vol. 3, no. 5, pp. 616–630, Oct. 2017, doi: [10.1016/j.eng.2017.05.015](https://doi.org/10.1016/j.eng.2017.05.015).
- [23] L. D. Xu, E. L. Xu, and L. Li, "Industry 4.0: State of the art and future trends," *Int. J. Prod. Res.*, vol. 56, no. 8, pp. 2941–2962, Apr. 2018, doi: [10.1080/00207543.2018.1444806](https://doi.org/10.1080/00207543.2018.1444806).
- [24] M. Kunath and H. Winkler, "Integrating the digital twin of the manufacturing system into a decision support system for improving the order management process," in *Proc. 51st CIRP Conf. Manuf. Syst. Integrating*, vol. 72, 2018, pp. 225–231, doi: [10.1016/j.procir.2018.03.192](https://doi.org/10.1016/j.procir.2018.03.192).
- [25] S. Boschert and R. Rosen, "Digital twin—The simulation aspect," in *Mechatronic Futures: Challenges and Solutions for Mechatronic Systems and Their Designers*, P. Hehenberger and D. Bradley, Eds., Cham, Switzerland: Springer, 2016, pp. 59–74, doi: [10.1007/978-3-319-32156-1](https://doi.org/10.1007/978-3-319-32156-1).
- [26] National Academies of Sciences, Engineering, and Medicine, *Foundational Research Gaps and Future Directions for Digital Twins*. Washington, DC, USA: The National Academies Press, 2024, doi: [10.17226/26894](https://doi.org/10.17226/26894).
- [27] D. L. V. Bossuyt et al., "The future of digital twin research and development," *J. Comput. Inf. Sci. Eng.*, vol. 25, no. 8, 2025, Art. no. 080801, doi: [10.1115/1.4068082](https://doi.org/10.1115/1.4068082).
- [28] W. Kritzing, M. Karner, G. Traar, J. Henjes, and W. Sihm, "Digital twin in manufacturing: A categorical literature review and classification," *IFAC-PapersOnLine*, vol. 51, no. 11, pp. 1016–1022, 2018, doi: [10.1016/j.ifacol.2018.08.474](https://doi.org/10.1016/j.ifacol.2018.08.474).
- [29] F. Tao, J. Cheng, Q. Qi, M. Zhang, H. Zhang, and F. Sui, "Digital twin-driven product design, manufacturing and service with big data," *Int. J. Adv. Manuf. Technol.*, vol. 94, nos. 9–12, pp. 3563–3576, Feb. 2018, doi: [10.1007/s00170-017-0233-1](https://doi.org/10.1007/s00170-017-0233-1).
- [30] A. Wooley, D. F. Silva, and J. Bitencourt, "When is a simulation a digital twin? A systematic literature review," *Manuf. Lett.*, vol. 35, pp. 940–951, Aug. 2023, doi: [10.1016/j.mfglet.2023.08.014](https://doi.org/10.1016/j.mfglet.2023.08.014).
- [31] K. M. Alam and A. E. Saddik, "C2PS: A digital twin architecture reference model for the cloud-based cyber-physical systems," *IEEE Access*, vol. 5, pp. 2050–2062, 2017, doi: [10.1109/ACCESS.2017.2657006](https://doi.org/10.1109/ACCESS.2017.2657006).
- [32] F. Tao and M. Zhang, "Digital twin shop-floor: A new shop-floor paradigm towards smart manufacturing," *IEEE Access*, vol. 5, pp. 20418–20427, 2017, doi: [10.1109/ACCESS.2017.2756069](https://doi.org/10.1109/ACCESS.2017.2756069).
- [33] C. Turner and J. Oyekan, "Manufacturing in the age of human-centric and sustainable industry 5.0: Application to holonic, flexible, reconfigurable and smart manufacturing systems," *Sustainability*, vol. 15, no. 13, p. 10169, Jun. 2023, doi: [10.3390/su151310169](https://doi.org/10.3390/su151310169).
- [34] F. Oetli, J. Schoeler, and J. Schilp, "Development of a method for evaluating the benefits of using a digital twin," in *Proc. Int. Conf. Elect., Comput., Commun. Mechatron. Eng.*, Feb. 2022, pp. 1–7, doi: [10.1109/ICECCME55909.2022.9988218](https://doi.org/10.1109/ICECCME55909.2022.9988218).
- [35] J. Barata and I. Kayser, "How will the digital twin shape the future of industry 5.0?" *Technovation*, vol. 134, Jun. 2024, Art. no. 103025, doi: [10.1016/j.technovation.2024.103025](https://doi.org/10.1016/j.technovation.2024.103025).
- [36] S. M. Sarathy, S. S. Nagaraja, E. Singh, E. Cenker, and A. Amer, "Review of life cycle assessments (LCA) for mobility powertrains," *Transp. Eng.*, vol. 10, Dec. 2022, Art. no. 100148, doi: [10.1016/j.treng.2022.100148](https://doi.org/10.1016/j.treng.2022.100148).
- [37] J. B. Guinée, R. Heijungs, G. Huppes, A. Zamagni, P. Masoni, R. Buonamici, T. Ekvall, and T. Rydberg, "Life cycle assessment: Past, present, and future," *Environ. Sci. Technol.*, vol. 45, no. 1, pp. 90–96, 2010.
- [38] L. Čuček, J. J. Klemeš, and Z. Kravanja, "A review of footprint analysis tools for monitoring impacts on sustainability," *J. Cleaner Prod.*, vol. 34, pp. 9–20, Oct. 2012, doi: [10.1016/j.jclepro.2012.02.036](https://doi.org/10.1016/j.jclepro.2012.02.036).
- [39] *Environmental Management—Life Cycle Assessment—Principles and Framework*, Standard ISO 14040:2006, 2006.
- [40] H. Hämäläinen and M. Ruusunen, "Identification of a supercritical fluid extraction process for modelling the energy consumption," *Energy*, vol. 252, Aug. 2022, Art. no. 124033, doi: [10.1016/j.energy.2022.124033](https://doi.org/10.1016/j.energy.2022.124033).
- [41] J. Mügge, J. Grosse Erdmann, T. Riedelsheimer, M. M. Manoury, S.-O. Smolka, S. Wichmann, and K. Lindow, "Empowering end-of-life vehicle decision making with cross-company data exchange and data sovereignty via Catena-X," *Sustainability*, vol. 15, no. 9, p. 7187, Apr. 2023, doi: [10.3390/su15097187](https://doi.org/10.3390/su15097187).
- [42] T. Riedelsheimer, L. Dorfhuber, and R. Stark, "User centered development of a digital twin concept with focus on sustainability in the clothing industry," *Proc. CIRP*, vol. 90, pp. 660–665, Jan. 2020, doi: [10.1016/j.procir.2020.01.123](https://doi.org/10.1016/j.procir.2020.01.123).
- [43] J. B. Oliveira, R. S. Lima, and J. A. B. Montevecchi, "Perspectives and relationships in supply chain simulation: A systematic literature review," *Simul. Model. Pract. Theory*, vol. 62, pp. 166–191, Mar. 2016, doi: [10.1016/j.simpat.2016.02.001](https://doi.org/10.1016/j.simpat.2016.02.001).
- [44] W. Trigueiro de Sousa Junior, J. A. Barra Montevecchi, R. D. Carvalho Miranda, and A. T. Campos, "Discrete simulation-based optimization methods for industrial engineering problems: A systematic literature review," *Comput. Ind. Eng.*, vol. 128, pp. 526–540, Feb. 2019, doi: [10.1016/j.cie.2018.12.073](https://doi.org/10.1016/j.cie.2018.12.073).
- [45] D. Tranfield, D. Denyer, and P. Smart, "Towards a methodology for developing evidence-informed management knowledge by means of systematic review," *Brit. J. Manage.*, vol. 14, no. 3, pp. 207–222, Sep. 2003, doi: [10.1111/1467-8551.00375](https://doi.org/10.1111/1467-8551.00375).
- [46] W. A. Junior, C. H. Santos, A. L. Romano, H. O. Salles, J. M. Palma, and R. T. A. Filho, "Integration of simulation and hierarchical approach to support lean implementation," *Int. J. Simul. Model.*, vol. 24, no. 1, pp. 111–122, Mar. 2025, doi: [10.2507/ijssimm24-1-719](https://doi.org/10.2507/ijssimm24-1-719).
- [47] A. P. Galvão Scheidegger, T. Fernandes Pereira, M. L. Moura de Oliveira, A. Banerjee, and J. A. Barra Montevecchi, "An introductory guide for hybrid simulation modelers on the primary simulation methods in industrial engineering identified through a systematic review of the literature," *Comput. Ind. Eng.*, vol. 124, pp. 474–492, Oct. 2018, doi: [10.1016/j.cie.2018.07.046](https://doi.org/10.1016/j.cie.2018.07.046).
- [48] D. Denyer, D. Tranfield, and J. E. van Aken, "Developing design propositions through research synthesis," *Org. Stud.*, vol. 29, no. 3, pp. 393–413, Mar. 2008, doi: [10.1177/0170840607088020](https://doi.org/10.1177/0170840607088020).
- [49] N. Bartie, L. Cobos-Becerra, M. Fröhling, R. Schlattmann, and M. Reuter, "Metallurgical infrastructure and technology criticality: The link between photovoltaics, sustainability, and the metals industry," *Mineral Econ.*, vol. 35, nos. 3–4, pp. 503–519, Dec. 2022, doi: [10.1007/s13563-022-00313-7](https://doi.org/10.1007/s13563-022-00313-7).
- [50] N. J. Bartie, Y. L. Cobos-Becerra, M. Fröhling, R. Schlattmann, and M. A. Reuter, "The resources, exergetic and environmental footprint of the silicon photovoltaic circular economy: Assessment and opportunities," *Resour., Conservation Recycling*, vol. 169, Jun. 2021, Art. no. 105516, doi: [10.1016/j.resconrec.2021.105516](https://doi.org/10.1016/j.resconrec.2021.105516).
- [51] X. Li, J. Cao, Z. Liu, and X. Luo, "Sustainable business model based on digital twin platform network: The inspiration from Haier's case study in China," *Sustainability*, vol. 12, no. 3, p. 936, Jan. 2020, doi: [10.3390/su12030936](https://doi.org/10.3390/su12030936).
- [52] L. Uhlenbrock, C. Jensch, M. Tegtmeier, and J. Strube, "Digital twin for extraction process design and operation," *Processes*, vol. 8, no. 7, p. 866, Jul. 2020, doi: [10.3390/pr8070866](https://doi.org/10.3390/pr8070866).
- [53] M. Rückert, S. Merkelbach, R. Alt, and K. Schmitz, "Online life cycle assessment for fluid power manufacturing systems – challenges and opportunities," *IFIP Adv. Inf. Commun. Technol.*, vol. 536, pp. 128–135, Jan. 2018, doi: [10.1007/978-3-319-99707-0_17](https://doi.org/10.1007/978-3-319-99707-0_17).
- [54] X. Vital, A. Krishnan, and H. Wyman, "Pushing the boundaries of LCA with AI integration and advanced life cycle inventories for semiconductor production," in *Proc. Electron. Goes Green*, Jun. 2024, pp. 1–4, doi: [10.23919/egg62010.2024.10631242](https://doi.org/10.23919/egg62010.2024.10631242).
- [55] A. O. Borjigin, J. Sresakoolchai, S. Kaewunruen, and J. Hammond, "Digital twin aided sustainability assessment of modern light rail infrastructures," *Frontiers Built Environ.*, vol. 8, pp. 1–12, Jul. 2022, doi: [10.3389/fbuil.2022.796388](https://doi.org/10.3389/fbuil.2022.796388).
- [56] S. Kaewunruen, S. Peng, and O. Phil-Ebosie, "Digital twin aided sustainability and vulnerability audit for subway stations," *Sustainability*, vol. 12, no. 19, p. 7873, Sep. 2020, doi: [10.3390/su12197873](https://doi.org/10.3390/su12197873).
- [57] S. Kaewunruen, M. AbdelHadi, M. Kongpuang, W. Pansuk, and A. M. Remennikov, "Digital twins for managing railway bridge maintenance, resilience, and climate change adaptation," *Sensors*, vol. 23, no. 1, p. 252, Dec. 2022, doi: [10.3390/s23010252](https://doi.org/10.3390/s23010252).
- [58] S. Kaewunruen, J. Sresakoolchai, W. Ma, and O. Phil-Ebosie, "Digital twin aided vulnerability assessment and risk-based maintenance planning of bridge infrastructures exposed to extreme conditions," *Sustainability*, vol. 13, no. 4, pp. 1–19, Feb. 2021, doi: [10.3390/su13042051](https://doi.org/10.3390/su13042051).
- [59] K. Lin, Y.-L. Xu, X. Lu, Z. Guan, and J. Li, "Digital twin-based life-cycle seismic performance assessment of a long-span cable-stayed bridge," *Bull. Earthq. Eng.*, vol. 21, no. 2, pp. 1203–1227, Jan. 2023, doi: [10.1007/s10518-022-01567-w](https://doi.org/10.1007/s10518-022-01567-w).
- [60] L. Rosafalco, S. Eftekhari Azam, S. Mariani, and A. Corigliano, "System identification via unscented Kalman filtering and model class selection," *ASCE-ASME J. Risk Uncertainty Eng. Syst., A, Civil Eng.*, vol. 10, no. 1, pp. 1–14, Mar. 2024, doi: [10.1061/ajrua6.rueng-1085](https://doi.org/10.1061/ajrua6.rueng-1085).

- [61] S. Kaewunruen and Q. Lian, "Digital twin aided sustainability-based lifecycle management for railway turnout systems," *J. Cleaner Prod.*, vol. 228, pp. 1537–1551, Aug. 2019, doi: [10.1016/j.jclepro.2019.04.156](https://doi.org/10.1016/j.jclepro.2019.04.156).
- [62] K. Slavkovic, A. Stephan, and G. Mulders, "A parametric approach to defining archetypes for an integrated material stocks and flows analysis and life cycle assessment of built stocks," in *Proc. Int. Conf. Architectural Sci. Assoc.*, 2022, pp. 55–65.
- [63] C. Chen, Z. Zhao, J. Xiao, and R. Tiong, "A conceptual framework for estimating building embodied carbon based on digital twin technology and life cycle assessment," *Sustainability*, vol. 13, no. 24, p. 13875, Dec. 2021, doi: [10.3390/su132413875](https://doi.org/10.3390/su132413875).
- [64] G. Artopoulos, P. Fokaides, V. Lysandrou, M. Deligiorgi, P. Sabatakos, and A. Agapiou, "Data-driven multi-scale study of historic urban environments by accessing Earth observation and non-destructive testing information via an HBIM-supported platform," *Int. J. Architectural Heritage*, vol. 18, no. 6, pp. 920–939, Jun. 2024, doi: [10.1080/15583058.2023.2199408](https://doi.org/10.1080/15583058.2023.2199408).
- [65] S. Wellsandt, C. Norden, R. Ahlers, S. Terzi, D. Cerri, D. Corti, and K.-D. Thoben, "Model-supported lifecycle analysis an approach for product-service systems," in *Proc. Int. Conf. Eng., Technol. Innov. (ICE/ITMC)*, Jun. 2017, pp. 858–864, doi: [10.1109/ICE.2017.8279973](https://doi.org/10.1109/ICE.2017.8279973).
- [66] C. Boje, Á. J. Hahn Menacho, A. Marvuglia, E. Benetto, S. Kubicki, T. Schaubroeck, and T. Navarrete Gutiérrez, "A framework using BIM and digital twins in facilitating LCSA for buildings," *J. Building Eng.*, vol. 76, Oct. 2023, Art. no. 107232, doi: [10.1016/j.jobte.2023.107232](https://doi.org/10.1016/j.jobte.2023.107232).
- [67] M. R. T. Williams, M. Reynolds, C. Parker, J. Chen, and K. Powell, "The impact of the electrification of buildings on the environment, economics, and housing affordability: A grid-response and life cycle assessment approach," *Digit. Chem. Eng.*, vol. 7, Jun. 2023, Art. no. 100086, doi: [10.1016/j.dche.2023.100086](https://doi.org/10.1016/j.dche.2023.100086).
- [68] A. C. Benedetti, C. Costantino, R. Gulli, and G. Predari, "The process of digitalization of the urban environment for the development of sustainable and circular cities: A case study of bologna, Italy," *Sustainability*, vol. 14, no. 21, p. 13740, Oct. 2022, doi: [10.3390/su142113740](https://doi.org/10.3390/su142113740).
- [69] M. Vaccari, G. Pannocchia, L. Tognotti, and M. Paci, "Rigorous simulation of geothermal power plants to evaluate environmental performance of alternative configurations," *Renew. Energy*, vol. 207, pp. 471–483, May 2023, doi: [10.1016/j.renene.2023.03.038](https://doi.org/10.1016/j.renene.2023.03.038).
- [70] H. Nikkiah, A. Di Maria, G. Granata, and B. Beykal, "Sustainable process design for lithium recovery from geothermal brines using chemical precipitation," *Resour. Conservation Recycling*, vol. 212, Jan. 2025, Art. no. 107980, doi: [10.1016/j.resconrec.2024.107980](https://doi.org/10.1016/j.resconrec.2024.107980).
- [71] G. Lupi, R. Accorsi, I. Battarra, and R. Manzini, "Life cycle assessment of reusable plastic containers throughout the fruit and vegetables supply chain," *IFAC-PapersOnLine*, vol. 58, no. 19, pp. 451–456, 2024, doi: [10.1016/j.ifacol.2024.09.253](https://doi.org/10.1016/j.ifacol.2024.09.253).
- [72] C. H. dos Santos, J. A. B. Montevechi, J. A. de Queiroz, R. de Carvalho Miranda, and F. Leal, "Decision support in productive processes through DesandABS in the digital twin era: A systematic literature review," *Int. J. Prod. Res.*, vol. 60, no. 8, pp. 2662–2681, Apr. 2022, doi: [10.1080/00207543.2021.1898691](https://doi.org/10.1080/00207543.2021.1898691).
- [73] M. Jašek and D. Kubečková, "The digital twin on a residential housing in the Czech republic on the BIM platform and its importance for the circular economy," *Structures*, vol. 78, Aug. 2025, Art. no. 109180, doi: [10.1016/j.istruc.2025.109180](https://doi.org/10.1016/j.istruc.2025.109180).
- [74] S. Weber, S. Forte, P. Savarino, T. Dickopf, and C. Apostolov, "Asset administration shell driven (Product) carbon footprint exchange with data management systems," in *Proc. IEEE 29th Int. Conf. Emerg. Technol. Factory Autom. (ETFA)*, Sep. 2024, pp. 1–6, doi: [10.1109/ETFA61755.2024.10710683](https://doi.org/10.1109/ETFA61755.2024.10710683).
- [75] B. Ben Daya, J.-F. Audy, and A. Lamghari, "Real-world data simulation comparing GHG emissions and operational performance of two sweeping systems," *Logistics*, vol. 8, no. 4, p. 120, Nov. 2024, doi: [10.3390/logistics8040120](https://doi.org/10.3390/logistics8040120).
- [76] E. Brusa, C. Gastaldi, C. Delprete, and L. Giorio, "A roadmap to the integration between systems engineering and circular design to develop sustainable industrial product," *IEEE Syst. J.*, vol. 18, no. 3, pp. 1693–1704, Sep. 2024, doi: [10.1109/JSYST.2024.3435025](https://doi.org/10.1109/JSYST.2024.3435025).
- [77] Z. Ullah, M. O. Aziz, G. Gruosso, R. Ullah, N. Ahmad, and A. Scacciantè, "Enhancing public building energy efficiency through digital twin modeling and iterative simulation: A sustainable approach," in *Proc. Int. Conf. Electr., Commun. Comput. Eng. (ICECCE)*, Oct. 2024, pp. 1–6, doi: [10.1109/iceccec63537.2024.10823633](https://doi.org/10.1109/iceccec63537.2024.10823633).
- [78] R. Accorsi, I. Battarra, B. Guidani, R. Manzini, M. Ronzoni, and L. Volpe, "Augmented spatial LCA for comparing reusable and recyclable food packaging containers networks," *J. Cleaner Prod.*, vol. 375, Nov. 2022, Art. no. 134027, doi: [10.1016/j.jclepro.2022.134027](https://doi.org/10.1016/j.jclepro.2022.134027).
- [79] H. Kim, "A real-time lifecycle analysis model with digital twin and novel control method for enhancing the environmental performance of electric/hybrid propulsion ships," *Ocean Eng.*, vol. 329, Jun. 2025, Art. no. 121195, doi: [10.1016/j.oceaneng.2025.121195](https://doi.org/10.1016/j.oceaneng.2025.121195).
- [80] P. Spudys, A. Jurelionis, and P. Fokaides, "Digitizing buildings sustainability assessment: Integrating energy audits, operational energy assessments, and life cycle assessments for enhanced building assessment," *Energy*, vol. 316, Feb. 2025, Art. no. 134429, doi: [10.1016/j.energy.2025.134429](https://doi.org/10.1016/j.energy.2025.134429).
- [81] G. Contini, F. Grandi, and M. Peruzzini, "Human-centric green design for automatic production lines: Using virtual and augmented reality to integrate industrial data and promote sustainability," *J. Ind. Inf. Integr.*, vol. 44, Mar. 2025, Art. no. 100801, doi: [10.1016/j.jii.2025.100801](https://doi.org/10.1016/j.jii.2025.100801).
- [82] M. Piron, E. Bovo, G. Lucchetta, and A. Manzardo, "Soft-sensors to drive manufacturing toward clean production: LCA based on digital twin," *J. Cleaner Prod.*, vol. 498, Mar. 2025, Art. no. 145192, doi: [10.1016/j.jclepro.2025.145192](https://doi.org/10.1016/j.jclepro.2025.145192).
- [83] J. Scholz, N. Klein, F. Köbller, and J. Fleischer, "Systematic digital twin-based development approach for holistic sustainable electric traction motors," *Sustainability*, vol. 17, no. 6, p. 2518, Mar. 2025, doi: [10.3390/su17062518](https://doi.org/10.3390/su17062518).
- [84] S. Wehking, A. Seegrün, T. Riedelsheimer, and K. Lindow, "Sustainable product lifecycle management—implementation of a digital twin of a biologically transformed product-ecosystem," *Proc. CIRP*, vol. 125, pp. 30–35, Jan. 2024, doi: [10.1016/j.procir.2024.08.006](https://doi.org/10.1016/j.procir.2024.08.006).
- [85] S. Aguacil Moreno, M. Loup, M. Lebre, L. Deschamps, J.-P. Bacher, and S. Duque Mahecha, "Virtual tours as effective complement to building information models in computer-aided facility management using Internet of Things," *Appl. Sci.*, vol. 14, no. 17, p. 7998, Sep. 2024, doi: [10.3390/app14177998](https://doi.org/10.3390/app14177998).
- [86] J. Zhao, X. Qu, Y. Wu, M. Fowler, and A. F. Burke, "Artificial intelligence-driven real-world battery diagnostics," *Energy AI*, vol. 18, Dec. 2024, Art. no. 100419, doi: [10.1016/j.egyai.2024.100419](https://doi.org/10.1016/j.egyai.2024.100419).
- [87] P. Dallasega, I. Kaushal, A. Revolti, and N. Miori, "Life cycle analysis for the concept design of a smart mobile factory (SMF) for infrastructure construction projects," in *IFIP Advances in Information and Communication Technology*, 2024, pp. 33–47.
- [88] J. V. S. do Amaral, C. H. dos Santos, J. A. B. Montevechi, and A. R. de Queiroz, "Energy digital twin applications: A review," *Renew. Sustain. Energy Rev.*, vol. 188, Dec. 2023, Art. no. 113891, doi: [10.1016/j.rser.2023.113891](https://doi.org/10.1016/j.rser.2023.113891).
- [89] C. H. D. Santos, J. A. de Queiroz, F. Leal, and J. A. B. Montevechi, "Use of simulation in the industry 4.0 context: Creation of a digital twin to optimise decision making on non-automated process," *J. Simul.*, vol. 16, no. 3, pp. 284–297, May 2022, doi: [10.1080/17477778.2020.1811172](https://doi.org/10.1080/17477778.2020.1811172).
- [90] A. Khodadadi and S. Lazarova-Molnar, "Data-driven extraction of simulation models for energy-oriented digital twins of manufacturing systems: An illustrative case study," in *Proc. Winter Simul. Conf. (WSC)*, Dec. 2024, pp. 1669–1680, doi: [10.1109/wsc63780.2024.10838974](https://doi.org/10.1109/wsc63780.2024.10838974).
- [91] B. C. Battissacco, W. A. Junior, C. H. dos Santos, and A. L. Romano, "Discrete event simulation-based digital twin: An approach focused on tactical, strategic, and operational decisions," *J. Simul.*, vol. 2025, pp. 1–19, Jul. 2025, doi: [10.1080/17477778.2025.2531045](https://doi.org/10.1080/17477778.2025.2531045).
- [92] X. Xu, G. Wang, H. Yan, L. Zhang, and X. Yao, "Deep-learning-enhanced digital twinning of complex composite structures and real-time mechanical interaction," *Composites Sci. Technol.*, vol. 241, Aug. 2023, Art. no. 110139, doi: [10.1016/j.compscitech.2023.110139](https://doi.org/10.1016/j.compscitech.2023.110139).
- [93] E. Wagner, S. Kegler, V. Gallina, T. Aigner, and J. Koller, "How industrial data meets information requirements of the digital product passport—The case of remanufacturing data," in *Proc. Electron. Goes Green*, Jun. 2024, pp. 1–7, doi: [10.23919/egg62010.2024.10631243](https://doi.org/10.23919/egg62010.2024.10631243).
- [94] B. Rodić, "Industry 4.0 and the new simulation modelling paradigm," *Organizacija*, vol. 50, no. 3, pp. 193–207, Aug. 2017, doi: [10.1515/orga-2017-0017](https://doi.org/10.1515/orga-2017-0017).
- [95] C. H. dos Santos, G. T. Gabriel, J. V. S. do Amaral, J. A. B. Montevechi, and J. A. de Queiroz, "Decision-making in a fast fashion company in the industry 4.0 era: A digital twin proposal to support operational planning," *Int. J. Adv. Manuf. Technol.*, vol. 116, nos. 5–6, pp. 1653–1666, Sep. 2021.

- [96] C. H. D. Santos, A. L. Romano, W. A. Junior, F. I. G. César, F. Ferraço, and I. Roza-Rojas, "State of the art and future directions of healthcare 4.0: A systematic literature review," *IEEE Access*, vol. 13, pp. 96469–96482, 2025, doi: [10.1109/ACCESS.2025.3575054](https://doi.org/10.1109/ACCESS.2025.3575054).
- [97] A. L. Romano, C. H. D. Santos, C. Koch, C. L. Ubeda, D. R. Fogaça, F. C. A. Santos, and V. Freitas, "Startups in the era of industry 4.0 and 5.0: Technologies, decisions, framing, and competences," *IEEE Access*, vol. 13, pp. 187042–187057, 2025, doi: [10.1109/ACCESS.2025.3627113](https://doi.org/10.1109/ACCESS.2025.3627113).
- [98] O. Hölck, S. Voges, K. F. Becker, E. Wagner, and M. Schneider-Ramelow, "Why should we share material data? The positive impact of public/private databases access through digital twins along the product value chain," in *Proc. Electronics Goes Green*, Jun. 2024, pp. 18–21, doi: [10.1007/978-3-031-84208-5_45](https://doi.org/10.1007/978-3-031-84208-5_45).



CARLOS HENRIQUE DOS SANTOS received the M.Sc., and Ph.D. degrees in production engineering from the Federal University of Itajubá (UNIFEI). He is currently a Researcher with the Center for Advanced Studies for Decision Aid (NEAAD), UNIFEI, and the Study and Research Group on Modeling and Simulation of Industrial Systems (GPMSSI), University of São Paulo (USP). He is a Professor of higher education with the Institute of Science and Technology, Federal University of Alfenas [Universidade Federal de Alfenas (UNIFAL)], Poços de Caldas, Brazil. He is the Coordinator of the Production Engineering Study Group for Integrated Solutions (GEPIS), UNIFAL. He selected as a short-term scholar to develop part of his Ph.D. research at North Carolina State University (NCSU), U.S., focusing on digital twins and machine learning. He has Black Belt certification in Six Sigma. His research focuses mainly addresses the areas of computer simulation and optimization, industry 4.0, digital twins, machine learning, and sustainability in production systems for goods and services.



ANA WOOLEY received the master's and Ph.D. degrees in industrial and systems engineering from Auburn University. She is currently an Assistant Professor with the Department of Industrial and Systems Engineering and Engineering Management (ISEEM), The University of Alabama in Huntsville (UAH). She is an expert in modeling and simulation of manufacturing systems, with a particular focus on automotive manufacturing systems. Her research focuses on digital twin development, smart manufacturing, simulation, and seamless integration of these technologies into manufacturing systems. She has collaborated with leading automotive companies and organizations, including Honda, MxD, and the Department of the Army. Her previous work includes developing digital twins and advanced simulations for automotive manufacturing lines, and conducting comparative analyses of the capabilities of digital twins and simulations in cyber-physical systems.



ANDRÉ LUIZ ROMANO received the bachelor's degree in economics and business administration, the Ph.D. degree in social sustainability and development from UAB, Lisbon, and the Ph.D. degree in production engineering from Unimep. He is a Researcher and a Professor with an academic trajectory and professional experience in diverse contexts. Currently, he is an Adjunct Professor with the Institute of Science and Technology, Federal University of Alfenas, Poços de Caldas, Minas Gerais Campus. He completed a Postdoctoral Fellowship in production engineering, where he studied sustainability in Brazilian organizations (UFSCar). He is also pursuing the Postdoctoral Fellowship focusing on digital transformation in Brazilian organizations (USP) within the production engineering program. He is an Associate Researcher with the Center for Environmental and Sustainability Research, NOVA University of Lisbon. His teaching experience spans various higher education institutions, where he also took on leadership roles, such as coordinating the business administration program. He contributes to the academic community as a scientific reviewer for journals and conferences. He is a member of the Production Engineering Study Group for Integrated Solutions (Unifal).

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